



Improved charging behaviors and flashover voltage of polypropylene film based on surface molecular structure modification

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ABSTRACT

The elevated temperature in operating capacitors speeds up the transportation of charges and causes the flashover to occur easily. Direct fluorination was used to modify the polypropylene film applied as capacitor dielectric, and behaviors of surface charging and flashover voltage at different temperatures were investigated. The surface morphology, functional groups and elemental composition of polypropylene film before and after modification treatment were characterized by scanning electron microscope (SEM), X-ray photoelectron spectroscopy (XPS) and energy dispersive x-ray analysis (EDAX), respectively. The surface charging behaviors were analyzed by the surface potential decay (SPD) method at temperature varying from 293 to 353 K. The flashover voltage was analyzed by the view of surface charge and trap characteristics. A thin layer of fluorination which changed the surface state of polypropylene film was formed, which can effectively inhibit the surface charge injection and accumulation at different temperatures. Shallow traps were introduced by the surface molecular structure modification, which was confirmed by the simulation of three-dimensional (3D) electrostatic potential and density of states (DOS). The modification of surface molecular structure, which suppresses the accumulation of surface charge leading to the inhibited distortion of electric field, effectively increased the flashover voltage at different temperatures.

1. Introduction

Power capacitors, employed to filter, provide voltage support, store energy, etc., are the crucial equipment in power system and the number is huge. The polypropylene film is widely used as the dielectric in power capacitors, considering the high dielectric strength and low dielectric loss [1,2]. Flashover is an important failure mechanism of power capacitors which can cause the expansion or even explosion of power capacitor [3]. Due to the electrode heating and dielectric loss, the temperature rise is inevitable when a capacitor operating [4–6]. The charging behaviors are related to surface charge which can cause the distortion of electric field and is an important factor to accelerate electrical aging of polymers [7–10]. The transportation of surface charge can be accelerated by the elevated temperature and then the anabatic electrical field distortion cause the flashover occur easily, which can decrease the lifespan of power capacitors [11–13]. For the operating capacitors with increased temperature, the suppression of the injection and transportation of charges in dielectric material is valuable both academically and industrially.

The electrical performance of the dielectric material is of great importance for the reliability of capacitors [14]. The injection of surface

charge is related to the surface state of dielectric materials. The defects (for example, fracture of chemical bonds and the folding of molecular) on the surface will form some energy level distribution in the band gap, namely the surface state [15]. When the dielectric material is under electrical stress, charge carriers can be captured by traps on the surface of dielectric and the trapped charges form the surface charge. Due to the inherent defects of the polypropylene film, premature dielectric failures occur frequently in power capacitors. Many studies have shown that direct fluorination can efficiently optimize the surface molecular structure of polymers and alter surface state [16–18], without changing inner property of polymers compared with nanocomposites [19–21]. The charging behaviors can be affected by direct fluorination for the formation of -(CFn) bonds on the polymer surface [22]. The effects of direct fluorination on charging behaviors and flashover voltage of polypropylene film especially at elevated temperature are not clear, which need further investigation.

In this paper, the surface molecular structure of polypropylene film was modified by direct fluorination with fluorine/nitrogen (F₂/N₂) mixture. The surface charging behaviors of polypropylene film with different contents of fluorine were investigated by surface potential decay (SPD) measurements at different temperatures. Besides, the trap

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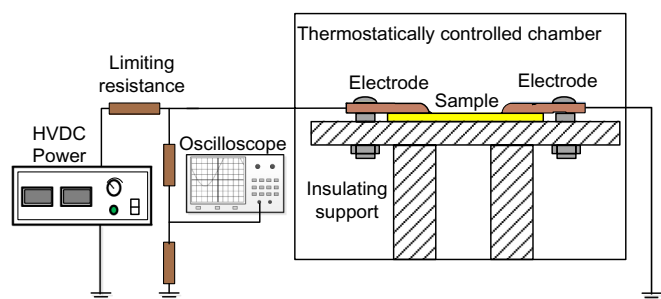


Fig. 1. The schematic diagram of flashover voltage measuring equipment.

distribution was analyzed. The flashover voltages at different temperatures were measured. The results show that the surface charging behavior is influenced by the surface modification and elevated temperature. The surface charge density decreases though direct fluorination, which is meaningful to the improved the flashover voltage.

2. Experimental

2.1. The surface molecular structure modification

The polypropylene film used in this paper is biaxially oriented, and the thickness of polypropylene film is 40 μm . In this paper, the surface molecular structure of polypropylene film was modified by direct

fluorination, which was conducted in a direct fluorination apparatus at 298 K with a F_2/N_2 mixture (vol% = 1:4) at 50 kPa. The modification time was controlled in 15 min to 60 min to obtain samples with different surface modification degrees. The specific surface modification process was detailed in our former research [23,24]. The temperature during modification was controlled by the temperature measuring regulator and the heating device.

2.2. Methods of sample characterization

The morphology features of cross section of samples were measured using the scanning electron microscope (SEM, Hitachi S-4800). The cross sections of samples were treated by the spray gold craft before the tests of SEM. The elemental composition of sample surface was analyzed using the energy dispersive x-ray analysis (EDAX, Ametek). Before the measurement of EDAX, the sample surface was treated by the spray gold craft.

The characterizations of chemical composition of samples were analyzed by X-ray photoelectron spectroscopy (XPS, Thermo Scientific ESCALAB 250Xi), using a $\text{Al K}\alpha$ X-ray source (1486.74 eV). The vacuum degree of testing chamber was kept at 5×10^{-10} mbar. The pass energy was set at 30.0 eV. The XPS spectra were analyzed in CasaXPS software (version 2.3.16, Casa Softw. Ltd). The binding energy (284.8 eV) of C–C was selected as the referenced binding energy, subtracting a Shirley background.

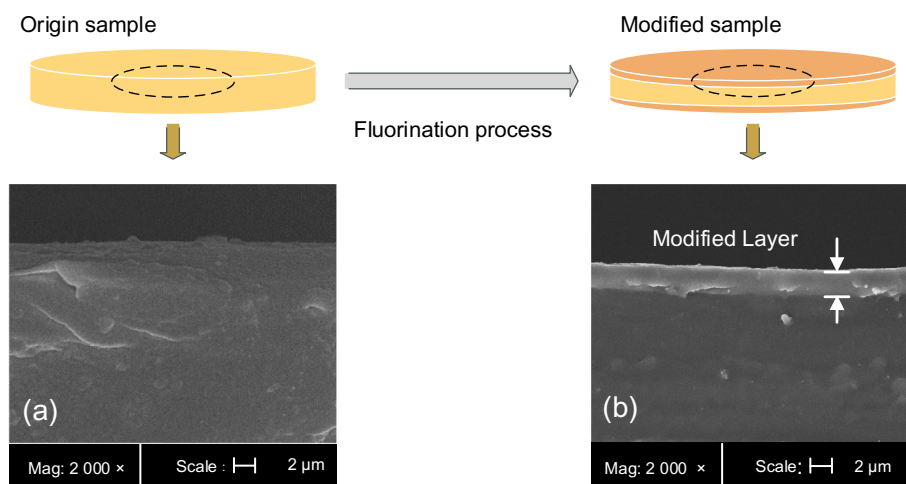


Fig. 2. SEM images of cross section of original sample (a) and cross section of modified sample (b).

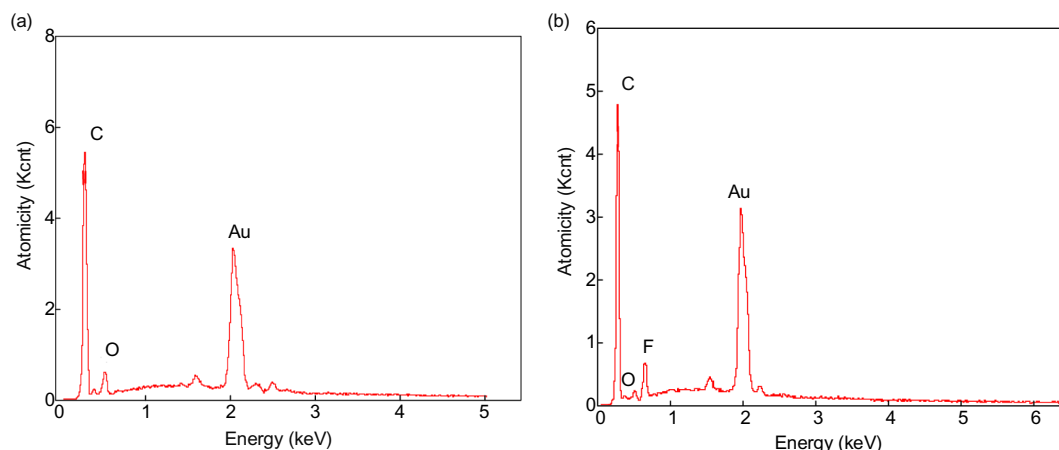


Fig. 3. The energy dispersive x-ray analysis of original sample (a) and modified sample (b).

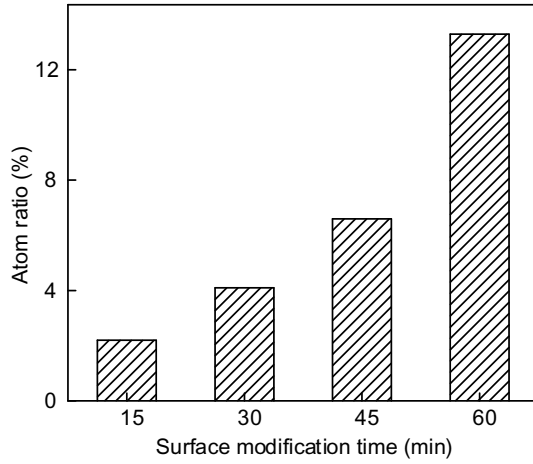


Fig. 4. Relative surface concentration of fluorine obtained by EDAX as a function of modification time.

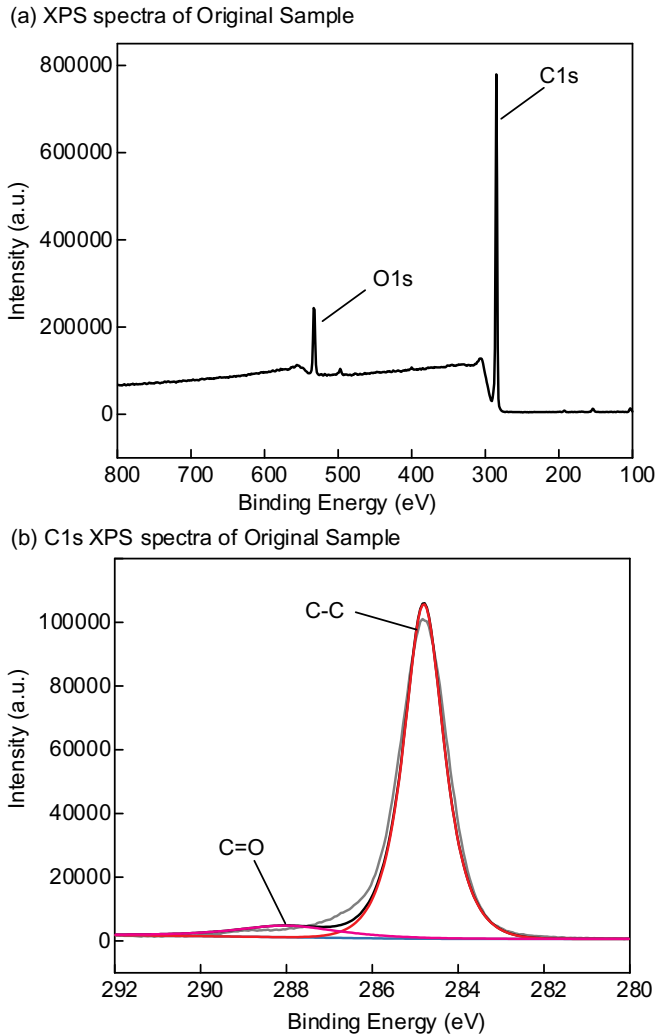


Fig. 5. (a) XPS spectra of original sample, (b) C1s XPS spectra of original sample.

2.3. Surface charging behaviors

The surface charging behaviors of original and modified samples at temperature of 293, 313, 333 and 353 K were initiated by the corona

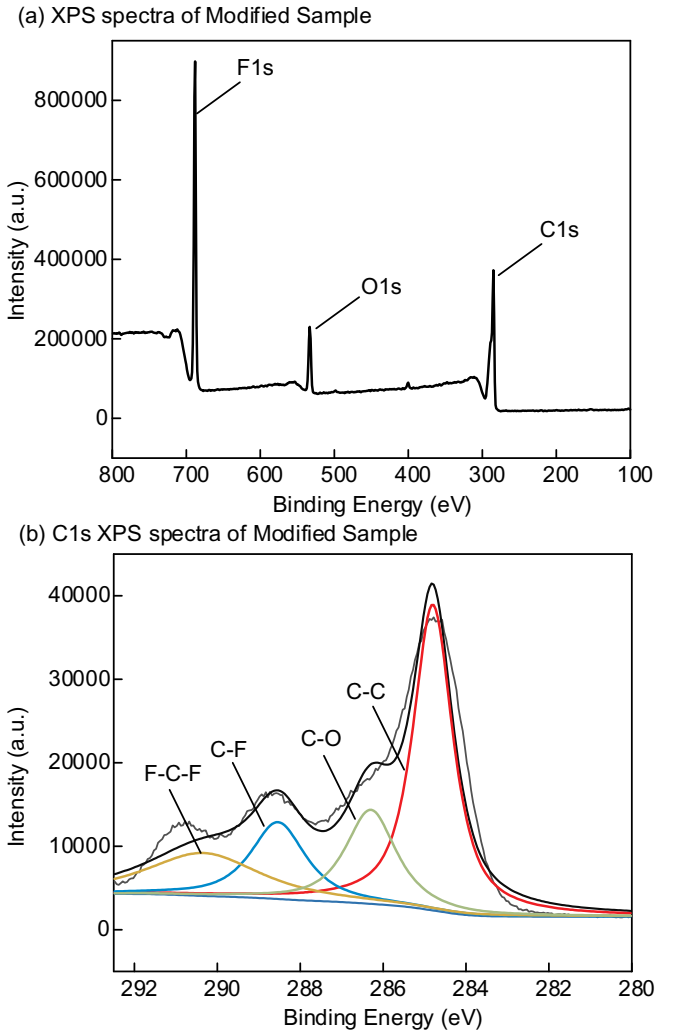


Fig. 6. (a) XPS spectra of modified sample, (b) C1s XPS spectra of modified sample.

charging of ± 5 kV DC voltage and measured by the Kelvin type probe (TREK-6000B-5C, TRECK). The experimental equipment and process were detailed in our previous work [25], except that the sample was placed on a heating platform which can heat the sample at the temperature varying from 293 to 353 K. Before the experiment, preheat the sample to the setting temperature on the heating platform for 10 min. The decay characteristics of surface charge can be represented by the SPD rate. The surface charge density can be obtained by the surface potential [26], as shown in the following expression.

$$\sigma = \frac{\epsilon_0 \epsilon_r}{L} V(t) \quad (1)$$

where σ is the surface charge density; ϵ_0 and ϵ_r are the vacuum and relative permittivity, respectively; L is the thickness of the sample and $V(t)$ is the surface potential at time t .

The transportation of surface charge, which can be represented by the trapping and de-trapping process of charge carrier, is closely related to the surface trap distribution that includes trap energy level and trap density [27]. Through the calculation of surface trap distribution, the trap characteristics can be obtained and then used to comprehend the transportation of surface charge. The expressions about the trap energy level E_t and trap density N_t were shown as follow [28].

$$E_t = kT \ln(\nu t) \quad (2)$$

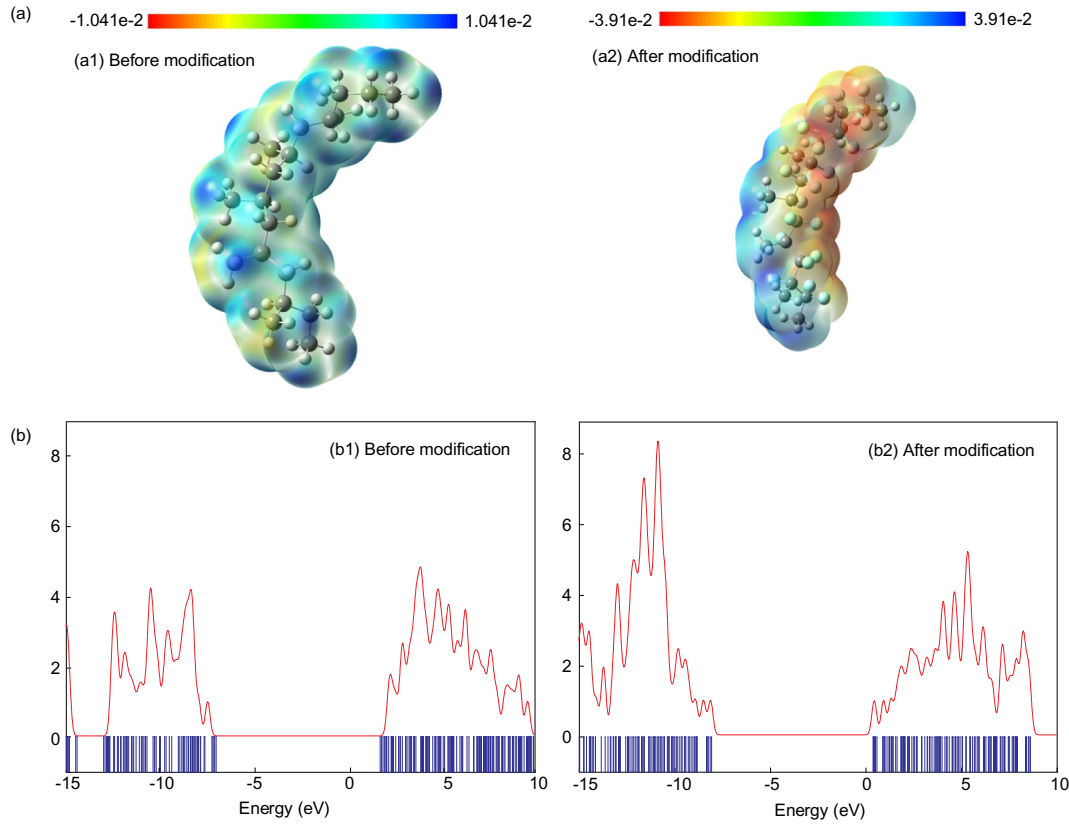


Fig. 7. (a) 3D potential distribution of the surface layer molecular chain of polypropylene before and after modification; (b) DOS spectrum and energy level of the surface layer of samples before and after modification.

Table 1

Surface potential decay rate of samples at different temperature.

Fluorination time (min)	Relative surface concentration of fluorine (at. %)	Surface potential decay rate (%)			
		293 K	313 K	333 K	353 K
0	0	2	25	59	74
15	2	4	28	63	85
30	4	4	32	66	86
45	7	5	40	74	92
60	13	4	30	70	90

$$N_i(E) = \frac{4\epsilon_0\epsilon_r}{eL^2kT} \left| t \frac{dV_s(t)}{dt} \right| \quad (3)$$

where e is electron volt; k is Boltzmann constant and ν is electronic escape frequency.

2.4. Flashover voltage measurement

The schematic diagram of flashover measuring equipment is shown in Fig. 1. The sample and electrodes were placed in a thermostatically controlled chamber, and then they were preheated to the setting temperature for 10 min before the measurement. Increasing the DC voltage at a speed of 0.5 kV/s until the flashover occurs. Experiments were performed at temperature 293, 313, 333 and 353 K. Every experiment was repeated for 20 times. The 2-parameter Weibull distributions is a most common method of statistical analysis. The value of voltage at 63.2% probability is the characteristic flashover voltage [29]. The Weibull distribution was employed to obtain the flashover voltage at the flashover probability of 63.2%.

3. Results and discussion

3.1. Sample characteristics

The fluorine can react with most polymers for its strong oxidizing property [30]. The SEM images of cross section of samples were obtained to study the morphology feature. The Fig. 2(a) is the SEM image of cross section of original sample, and the Fig. 2(b) is the SEM image of cross section of the sample modified for 45 min. For the modified sample, a thin layer, which differ with inner section, was formed on sample surface. The thickness of samples fluorinated for 15, 30, 45 and 60 min were 1.12 ± 0.16 , 2.10 ± 0.21 , 2.79 ± 0.18 and 3.01 ± 0.25 μm , respectively. Besides, the surface of modified sample was smoother than the surface of original sample. The behaviors of surface charging are affected by the trap distribution of the dielectric material, which is closely associated with the surface state of the dielectric material. The change of surface morphology and the formed thin layer are the main reasons for the change of surface state of samples.

To confirm the existence of fluorine, the EDAX spectrometry of sample surface was obtained to analyze the elemental composition. The surface elemental compositions of original sample and modified sample are shown in Fig. 3. For the original sample, only carbon, oxygen, gold element can be obtained. Incidentally, the gold element is introduced by the spray gold craft, which is essential for the observation. For the modified sample, except for the carbon, oxygen and gold element, fluorine element can also be found. Results in Fig. 2 and Fig. 3 confirmed that surface modification by direct fluorination introduces the fluorine to the sample and forms a fluorination layer at the sample surface. The atom ratio can be regarded as the degree of surface modification. The atom ratio is the proportion of one element in the total elements in the measuring range. The relative surface concentration of

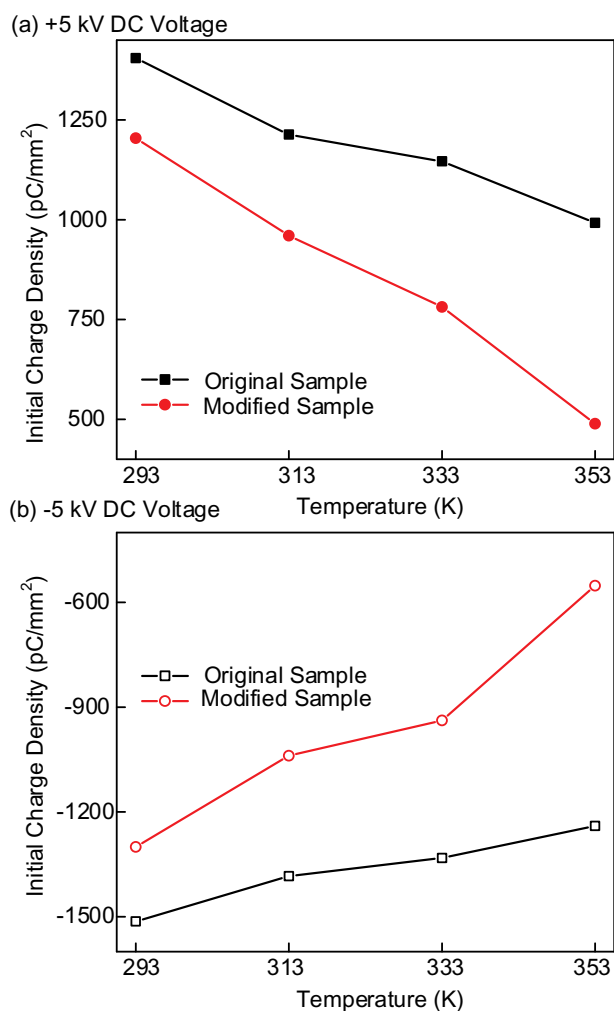


Fig. 8. The initial surface charge density at different temperature under positive corona charging voltage (a) and negative corona charging voltage (b).

fluorine obtained by EDAX as a function of modification time is shown in Fig. 4. The atom ratio of fluorine element increase with the increasing modification time.

In this paper, for investigating the existential state of fluorine in polypropylene film, the XPS was employed to test the surface chemical composition of samples. The XPS spectra of original sample is shown in Fig. 5. Only carbon and oxygen elements can be obtained in the XPS spectra of original sample. The XPS spectra of modified sample is shown in Fig. 6(a). Carbon, oxygen and fluorine elements were obtained in the modified sample, indicating fluorine element were introduced in modified sample. The C1s XPS spectra of modified sample is shown in Fig. 6(b). There are four peaks in the C1s XPS spectra, including C–C bonds (284.8 eV), C–F bonds (288.5 eV), C–F₂ bonds (290.3 eV) and C–O bonds (286.3 eV). For the modified samples, the hydrogen in -(CH_n) groups were replaced by fluorine. When the fluorine composite attached to the polypropylene surface, the chemical reaction took place and the C–H bonds were replaced by C–F [31]. The C–F bonds have a higher stability for the higher bonds energy of C–F bonds comparing with C–H bonds [32,33]. The C–O and C=O bonds may be caused by impurity. It can be concluded that the fluorine in the fluorination layer in the form of C–F bonds on the sample surface after surface molecular structure modification.

3.2. Three-dimensional (3D) electrostatic potential and density of states (DOS) characteristics

The trap distribution is closely associated with the electrostatic potential in polymer. Fig. 7(a) shows the 3D electrostatic potential in the molecular chain of polypropylene, which was employed to study the effect of the introduced fluorine on the electrostatic potential. In Fig. 7(a), the warm colors represent the negative potential and the cold colors represent the positive potential, respectively. The orbital electron is deviated in molecular chains for the difference of electronegativity between the carbon and hydrogen in polypropylene. For the modified samples, both the warm colors and the cold colors were deepened, meaning that both the electrons and holes can be captured much more easily.

Fig. 7(b) exhibits the energy level distribution and DOS spectrum in surface layer of original sample and modified sample. There are many traps formed in amorphous region in polypropylene film, which is a kind of semi-crystalline polymers [34,35]. The molecular chains arrange closely and the interaction force between molecular chains is strong in crystalline region. While, the molecular chains arrange loosely and disordered, and there are branched chain, shorter molecules in amorphous region. These structure defects in amorphous region introduce additional energy level in band gap. Broken chemical bonds or folding molecular chain on sample surface cause the distortion or interrupt of the inner potential field. The arrangement of atoms on the surface tends to be adjusted for the stabilization of whole system. Therefore, many energy traps are generated on the sample surface. In the original sample surface, the lowest unoccupied molecular orbital (LUMO) is 2.24 eV, and the highest occupied molecular orbital (HOMO) is -7.43 eV. Therefore, the energy gap of the original sample is 9.67 eV. For the modified sample, the LUMO is 1.6 eV and the HOMO is -7.49 eV. The energy gap of modified sample is 8.09 eV. It can be found that the LUMO of modified sample is lower than that of original sample. The formation of -(CF_n) bonds generates some new electron traps on the sample surface [36]. Direct fluorination can introduce shallow traps in the sample surface, which can be inferred by the newly generated electron traps. The electronic potential and DOS spectrum were employed to analyze the influence of fluorine replacement on the surface state of samples, so the ideal molecular chain of polypropylene was used in this paper.

3.3. Surface charging

Charges are generated around the needle electrode by corona charging, and are forced to the sample surface under the electric field between the needle electrode and ground electrode. The SPD process was measured immediately after the finish of corona charging with recording time for 1200 s. The SPD rate can be obtain by the expression as follow [37].

$$P_d = \frac{V(t_0) - V(t_{end})}{V(t_0)} \times 100\% \quad (4)$$

where P_d is the potential decay rate; $V(t_0)$ is the initial surface potential and $V(t_{end})$ is the surface potential at time t_{end} . In this paper, t_{end} is set for 1200 s.

There are two factors, temperature and the degree of modification which can be represented by atom ratio of fluorine, affects the surface potential decay rate. Charges trapped in energy traps can obtain more energy and de-trap much easier at elevated temperature, which lead the SPD rate increase with the increasing temperature. For the sample with 0% atom ratio of fluorine, namely original sample, in Table 1, the SPD rate increases from 2.4% to 74% when temperature varying from 293 to 353 K. The surface potential decay rate of sample with 7.59% atom ratio of fluorine is higher than other samples. This paper mainly took the sample with 7.59% atom ratio of fluorine, namely the sample modified for 45 min, to analyze the relationship between the surface

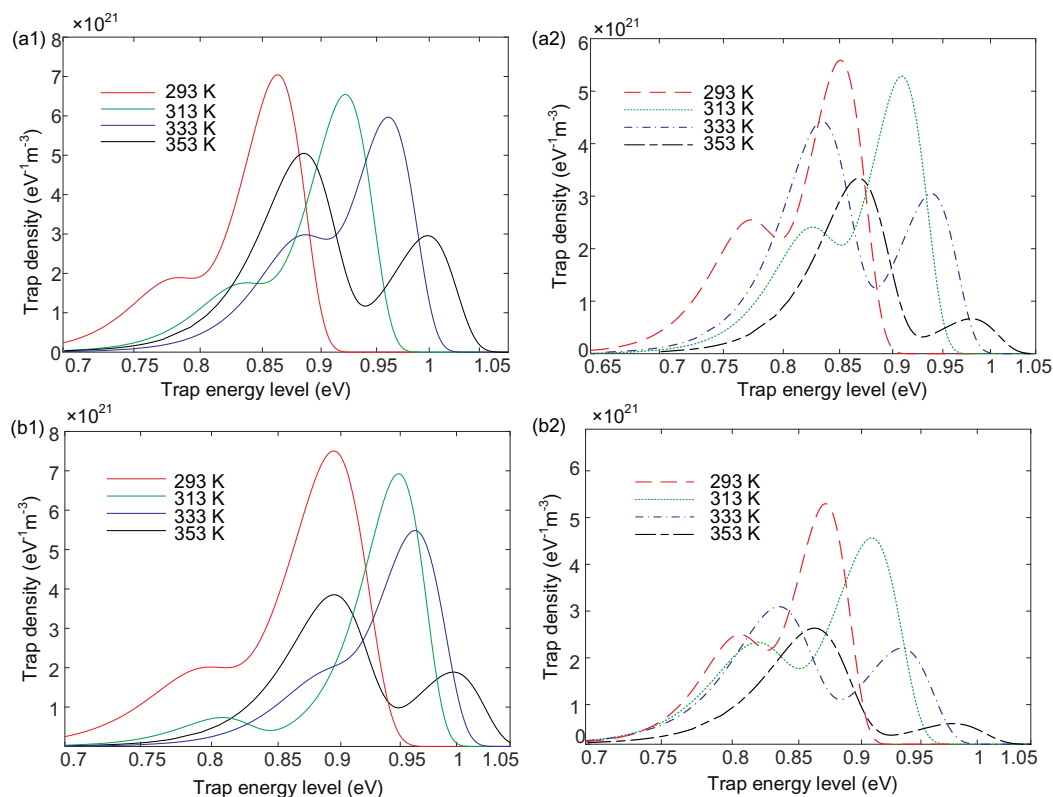


Fig. 9. Surface trap distribution of samples under 5 kV corona voltage; electron traps of original sample (a1), electron traps of modified sample (a2), hole traps of original sample (b1), hole traps of modified sample (b2).

molecular modification and temperature. The surface molecular modification changes the surface state of the sample, which was presented by the shallow traps. The charges de-trap much easier from the shallow traps which cause the increased decay rate.

The initial surface charge density (ISCD) reflects the injection and accumulation of surface charge during the corona charging process, which is related with the surface state of samples. Fig. 8 shows the values of ISCD varying with different temperatures. The modified sample represents the sample with 7.59% atom ratio of fluorine. In Fig. 8(a), the ISCD of the original sample decreases with temperature when under positive DC voltage. Charges induced from needle electrode during corona charging process get to the sample surface under the electric field and are captured in traps on the sample surface. For one thing, the electric field excited by the trapped charges decreases the electric field from the needle electrode to the ground electrode, which restrains the injection of charge. Besides, the dissipation process of charges begins during the corona charging. The dynamic equilibrium can be established during the corona charging process, and a stable number of charges accumulated on the sample surface, which can be represented by the initial surface charge. Much more energies can be obtained by charges at higher temperature, which make charges de-trap much easier and quicker. Therefore, in a macroscopic scale, the ISCD declines with the increasing temperature. The ISCD of modified sample decreases with temperature, and the trend is more obvious than that of original sample. Charges transport to the bulk of sample by trapping and de-trapping process. Many charges trapped in deep traps on surface and in the bulk of original sample, a high ISCD is measured by the electrostatic voltmeter. Whereas, there is a fluorination layer formed on the surface of modified samples, which make it easier to de-trap from shallow traps for charges. Therefore, the migration of charges from the sample surface into the bulk of sample was suppressed. As a result, the

ISCD of modified sample is lower than the ISCD of original sample. The data in Fig. 8 (a) indicate that the ISCD of modified samples is lower than the ISCD of original sample at different temperatures, which implies the suppression effects of surface charge injection by surface modification still exists even at higher temperature. In Fig. 8(b), similar rules are obtained for original sample and modified sample when the corona charging voltage is negative, except that the ISCD is higher. Because the charge quantity around a needle electrode induced by negative corona charging voltage is larger. Besides, the difference between the mobility of the hole carriers and electrons affects the ISCD.

3.4. Surface trap distribution

The results of surface trap distribution obtained by SPD are shown in Fig. 9. The trap distribution shows two characteristic peaks of trap density, namely deep trap level (DTL) and shallow trap level (STL) which are relative. The trap energy level of DTL is higher, while the trap energy level of STL is lower. Carriers in deep traps need more energy to de-trap than that in shallow traps. The trap levels of both hole trap and electron trap in original and modified samples trend to increase with temperature. The elevated temperature enhances the energy of charges and the probability of de-trapping from deeper traps. Comparing the Fig. 9(a1) with Fig. 9(a2), or the Fig. 9 (b1) with Fig. 9 (b2), the existence of fluorination layer lowers the STL and DTL of samples at different temperatures. This is the fundamental cause of the inhibition effects of direct fluorination on the surface charges. For the original sample, the density of deep traps is much higher than the density of shallow traps at lower temperature, which indicates that there are much more deep traps in original sample. Therefore, charges are easily accumulated in the original sample. The elevated temperature causes more charges de-trap from shallow traps, therefore, the density of deep

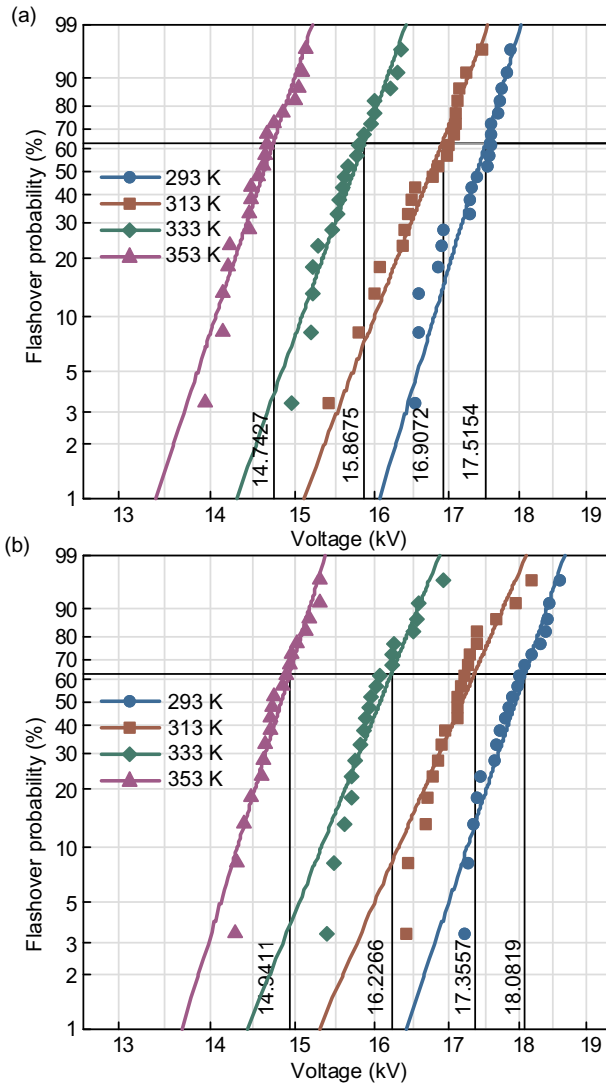


Fig. 10. Flashover voltage at different temperatures of original sample (a) and modified sample (b).

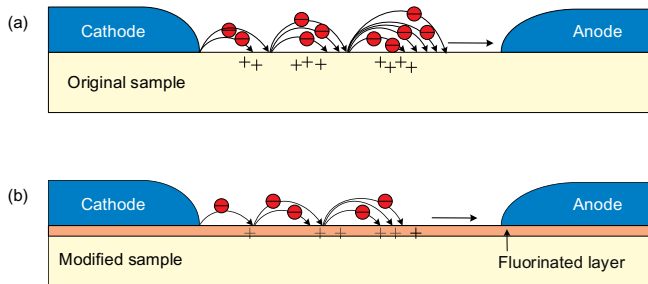


Fig. 11. The schematic model of flashover voltage of original sample (a) and modified sample (b).

traps decreases and the density of shallow traps increases at a higher temperature. For the modified sample, the number of hole traps is impaired comparing with the electron traps, which implies that the direct fluorination greatly affects the hole traps. Furthermore, the shallow traps, no matter hole or electron traps, introduced by direct fluorination, meanwhile, the deep traps were suppressed. The trap distribution obtained by SPD confirmed the results of the simulation in Section 3.2.

3.5. Flashover voltage

The flashover voltage at the flashover probability of 63.2% was obtained. Fig. 10 demonstrates the flashover voltages of original sample and modified sample. In Fig. 10(a), for the original sample, the surface flashover voltage presented a decrease trend with the increasing temperature. Electrons obtained high kinetic energy under the effects of elevated temperature. The hot electron can surmount the potential barrier and the thermionic emission occurs easier. Therefore, initial electrons that participate in flashover process increased. Besides, the increased kinetic energy speeds up the impact ionization and the development of the electron avalanche. The flashover voltage of modified sample presents a similar trend that decreases with increasing temperature. Furthermore, the flashover voltages are elevated by direct fluorination at all temperatures.

The schematic model of flashover voltage of original sample and modified sample is shown in Fig. 11. For the modified sample, the impact ionization is restrained, which makes the secondary electron emission harder to occur and the formation of electron avalanche is restrained. Besides, the direct fluorination leads to surface charge decrease which suppressed the electrical field distortion, then the initiation probability of flashover decreases. The ISCD of samples as mentioned above confirmed this conclusion. For another, the transportation of charges on sample surface is affected by the trap energy level. The charges can be trapped by traps during the process of directional migration under high electric field, then migrate at conduction band by the effects of thermal excitation, and then captured by other traps. The trap level of modified sample is shallowed, which is helpful to increase the possibility of the trapping and de-trapping. While, the decrease of trap density has opposite effect. The comprehensive effects of trap energy and trap density may suppress the transportation along the surface.

4. Conclusions

Direct fluorination was employed as a modify method to modify the surface molecular structure of polypropylene film. The thin fluorination layer was formed on the sample surface and the $-(CF_n)$ bonds were formed, which changed the surface state of samples. The variation of surface structure and chemical composition introduced shallow traps on the sample surface, which increased the SPD rate and suppressed the surface charge accumulation at temperature varying from 293 to 353 K. The STL and DTL obtained by SPD were shallowed at all temperature conditions. The surface flashover voltage was increased for the less distortion of electric field induced by surface charges accumulation. All of above results verify the direct fluorination is a promising candidate in improving the electric property of dielectric material for power capacitor.

Acknowledgements

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